

Household Sorting and the Impact of Mobility Costs

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Motivation – Housing and Barriers to Mobility

- Housing is important
 - Imputed homeowner rents contribute around 9% of U.S. GDP ($\approx$ \$2 trillion).
 - Home equity comprises 45% of median homeowner's net worth.
- Mobility costs are high: estimates in the literature well into the \$300,000s.

Bishop (2008); Bayer, Keohane, Timmins (2009); Kennan and Walker (2011)
- High mobility costs create potential for misallocation.

Venti and Wise (1984); Glaeser and Luttmer (2003); Hsieh and Moretti (2019)
- Realtor fees a key barrier to mobility

What we do

What is the impact of mobility barriers on household and realtor welfare?

- ① Develop a novel identification strategy
- ② Estimate moving costs
 - financial and non-financial costs
- ③ Quantify heterogeneity in effects on mobility
- ④ Measure mean impact of counterfactually reducing moving costs
 - Halving Realtor fees substantially increases mobility

Contributions

- Discrete choice models of housing/sorting:

- Bayer, Ferreira, McMillan (2007); Galiani, Murphy, Pantano (2015); Bayer, McMillan, Murphy, Timmins (2016); Murphy (2018); Diamond, McQuade, Quian (2019); Almagro and Dominguez-lino (2025)

Our contribution: Adapt frontier methods to setting of mobility withr Realtor fees.

- Estimates of mobility costs and mobility cost reduction policies:

- Bishop (2008); Bayer, Keohane, Timmins (2009); Kennan and Walker (2011); Galiani, Murphy, Pantano (2015); Bergman et al. (2024)

Our contribution: Show impact of mobility costs on mobility and sorting behavior

- Behavior of Realtors and their impact on the housing market:

- Hsieh and Moretti (2003); Levitt and Syverson (2008); Jia and Pathak (2010); Barwick and Pathak (2015); Barwick, Pathak, Wong (2017); Kim (2025); Buchak, Matvos, Piskorski, Seru (2025)

Our contribution: Show that Realtor fee reductions have large welfare effects.

Outline

- 1 Mobility Costs and Realtor Fees
- 2 Identification Challenge
- 3 Overview of Research Design
- 4 Data
- 5 Model and Estimation
- 6 Results
- 7 Conclusion

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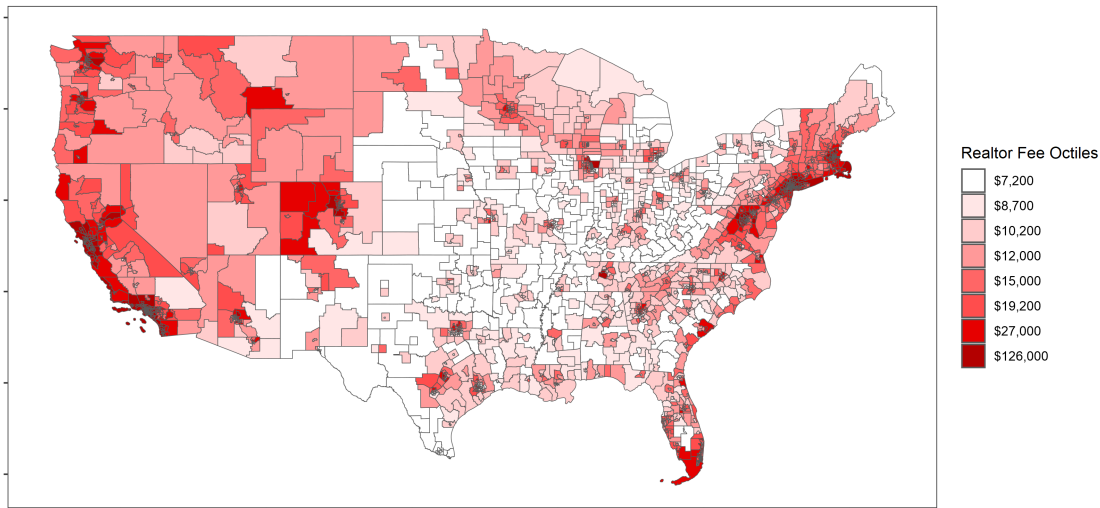
Mobility Costs

- Estimated in \$000,000s
 - Bishop(2008); Bayer, Keohane, Timmins (2009); Kennan and Walker (2011)
 - Somewhat difficult to interpret, but very large
- Mobility cost reduction policies:
 - Galiani, Murphy, Pantano (2015)
 - Bergman et al. (2024)

Mobility Costs – Realtor Fees

- Rates are large and spatially invariant at 6%
 - Realtor Fee Rate: RFR
- Absolute amounts large and vary due to spatially varying prices
 - Realtor Fee: $RFR \cdot P_j$
- Annualized Realtor fees are 10-15% of total cost of housing
- 2024 NAR settlement
 - Pay \$418 million
 - Restructure commission basis (with no effect)

Mobility Costs – Realtor Fees – 6% House Prices



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Identification Challenge

Consider a regression of mobility on neighborhood characteristics and moving costs:

$$move_{ij} = \underbrace{v(X_k, \xi_k, UCR \cdot P_k)}_{\text{value of moving to } k} - \underbrace{v(X_j, \xi_j, UCR \cdot P_j)}_{\text{value of staying in } j} - \underbrace{MC(RFR \cdot P_j)}_{\text{moving costs from } j} + \epsilon_{ijk}$$

- k (j) denotes best destination (existing) neighborhood
- X (ξ) denotes observable (unobservable) neighborhood characteristics
- UCR denotes user-cost rate.
 - maps asset price into annualized cost
- P denotes price.
- RFR denotes Realtor fee rate

Identification Challenge

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- Fails both conditions for identification!
- Rank Condition:
 - $RFR \cdot P_j$ is perfectly colinear with $UCR \cdot P_j$
- Order Condition:
 - Price, P_j is correlated with unobserved term ξ_j

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High-level Overview of Research Design

- Secure identification via additional data (and some structure)
 - Add data on spatially varying UCR_j (Bishop, et al. (2025))
 - Use location decision conditional on moving (Galiani et al. 2015)
- Household decisions and identification (BMMT 2016)
 - Neighborhood choice, given moving, identifies neighborhood prefs
 - Frequency of moving identifies mobility costs
- Household decisions and identification (Schiraldi 2011)
 - Stocks and flows identify neighborhood prefs and mobility costs

Overview of Research Design

$$move_{ij} = \underbrace{v(X_k, \xi_k, UCR \cdot P_k)}_{v_k = \text{value of moving to } k} - \underbrace{v(X_j, \xi_j, UCR \cdot P_j)}_{v_j = \text{value of staying in } j} - \underbrace{MC(RFR \cdot P_j)}_{\text{moving costs from } j} + \epsilon_{ijk}$$

$$move_{ij} = (v_k - v_j) - MC(RFR \cdot P_j) + \epsilon_{ijk}$$

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 - Hotz Miller (1993): $(v_k - v_j)$ identified from neighborhood choice probabilities

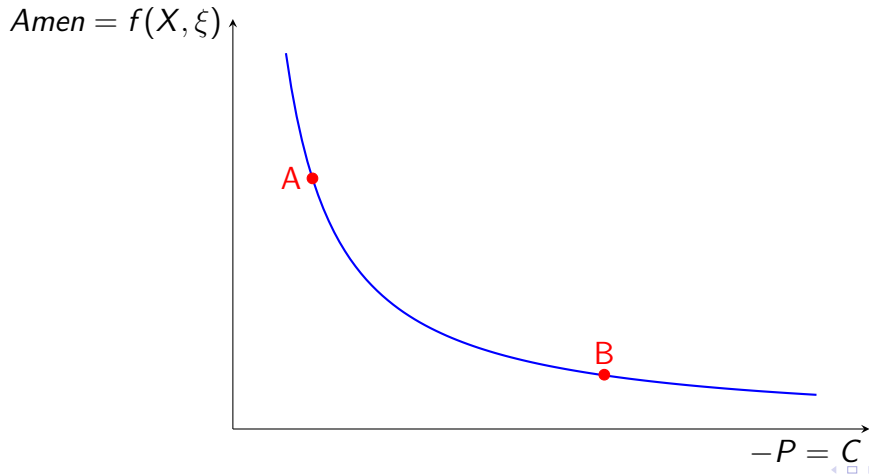
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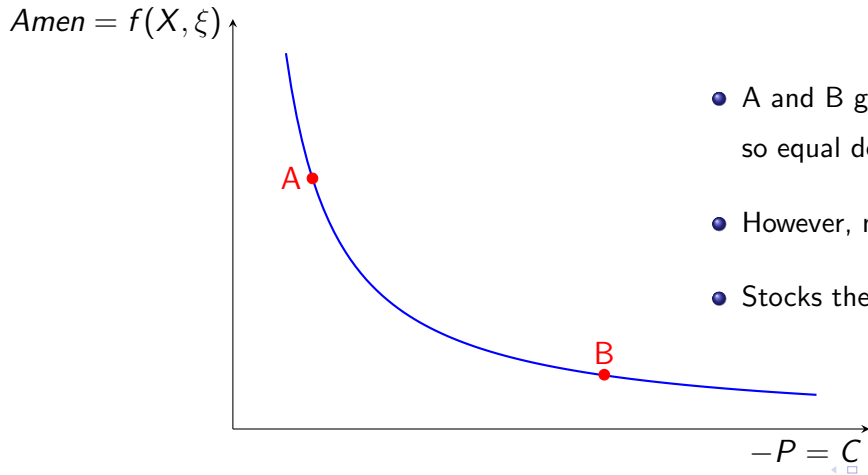
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- If we can observe and include $(v_k - v_j)$, this solves order and rank problems!
 - Hotz Miller (1993): $(v_k - v_j)$ identified from neighborhood choice probabilities
- But, what does it mean to vary P_j , holding v_j constant?

Indifference Curve: $U(\text{Amen}, C) = U(\text{Amen}, -P) = \delta$



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- A and B give the same flow utility, so equal demand|move
- However, mobility out of A lower
- Stocks the same, but flows differ

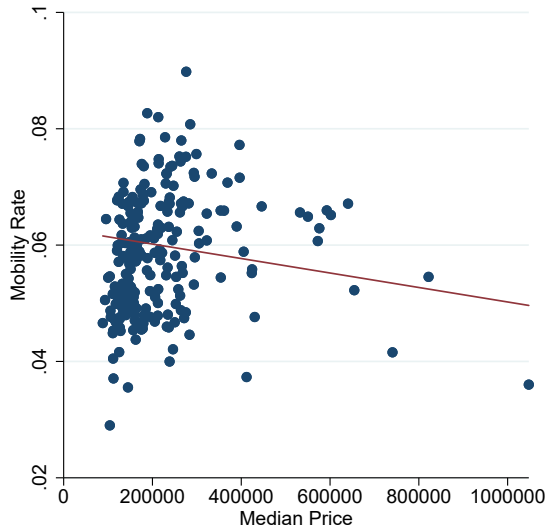
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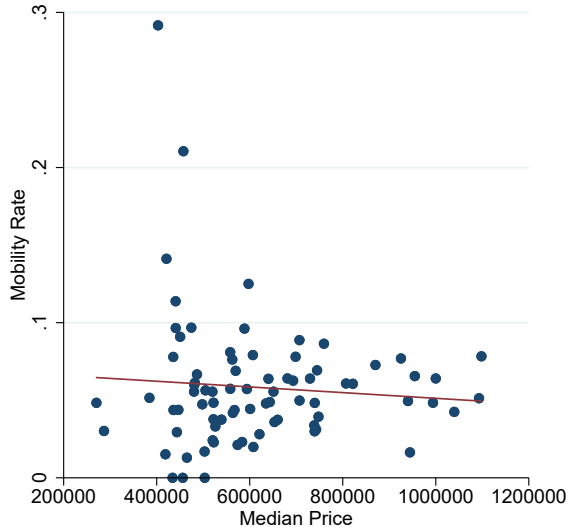
Data

- ACS – focus on 2016-2019
- For over 3 million homeowners, observe
 - Current location, i.e., PUMA
 - Whether moved in the last year
 - Prior location
 - Tenure
 - Rich demographics
 - Neighborhood mean attributes, including price/value
- UCR data from Bishop et al. (2025)

Data – Mobility Rates and Prices by MSA



Data – Mobility Rates and Prices by PUMA in LA



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Model Overview

- Model of *within-city* migration
 - 225 MSAs
- Households have observable attributes, Z_i .
- Households maximize lifetime utility by choosing neighborhood (PUMA)
 - Households can stay, move within neighborhood, or change neighborhood.
- Households that move incur mobility costs: psychological and financial

Model and Estimation

- The full utility that household i which begins a period in neighborhood h_i , receives from choosing alternative/neighborhood j is given by:

$$v_{ij} - MC_{ih_i} 1_{[j \neq J+1]} + \epsilon_{ij}$$

- v_{ij} flexibly captures observable (X_j, UC_j) , and unobservable (ξ_j) attributes of neighborhoods and observable attributes of households (Z_i) .
- Moving from neighborhood h_i to neighborhood j reduces utility by MC_{ih_i} .
 - MC_{ih_i} is not a function of the destination neighborhood, j .
- $J + 1$ denotes not moving and staying in existing neighborhood, h_i

Model and Estimation

- Household-specific unobserved heterogeneity in this term comes through ϵ_{ij} .
 - Assume ϵ_{ij} is distributed i.i.d., Type 1 Extreme Value.
- The probability that household i chooses neighborhood j is given by:

$$P_{ij} = \frac{\exp(v_{ij} - MC_{ih_i} 1_{[j \neq J+1]})}{\exp(v_{ih_i}) + \sum_{k \neq J+1} \exp(v_{ik} - MC_{hi})}$$

Model and Estimation

- The model can be estimated in two stages.
- Step 1: Focus on movers. The probability of choosing j , given $j \neq J + 1$ is:

$$P_{ij|d_i \neq J+1} = \frac{\exp(v_{ij} - MC_{ih_i})}{\sum_{k \neq J+1} \exp(v_{ik} - MC_{ih_i})} = \frac{\exp(v_{ij})}{\sum_{k \neq J+1} \exp(v_{ik})}$$

- Assumption that MC doesn't vary with destination causes MC to drop out.
- Now a simple conditional logit model, inverting shares yields v_{ij} terms.

$$v_{ij} = \log(P_{ij|d_i \neq J+1}) - \log(P_{0|d_i \neq J+1})$$

- Neighborhood choice probabilities, P_{ij} , can be recovered from data

Model and Estimation

- Step 2: Use the move/stay decision for all households to estimate MC .
- The mobility probability is given by:

$$P_i^{move} = \frac{\sum_{k \neq J+1} \exp(v_{ik} - MC_{ih_i})}{\exp(v_{ih_i}) + \sum_{k \neq J+1} \exp(v_{ik} - MC_{ih_i})}$$

where v_{ik} and v_{ih_i} are known/estimated from first stage

Model and Estimation

- Parameterize moving costs as $MC_{ij} = \gamma_{0i} + \gamma_{1i}RFR \cdot P_j$
- Estimate γ using a binary logit model where the net utility of moving from j is :

$$move_{ij} = \log\left(\sum_{k \neq J+1} \exp(v_{ik})\right) - v_{ij} - (\gamma_{0i} + \gamma_{1i}RFR \cdot P_j)$$

Model and Estimation

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$$move_{ij} = \underbrace{\log\left(\sum_{k \neq J+1} \exp(v_k)\right)}_{\text{utility of living in max } k} - \underbrace{v_j}_{\text{utility of living in } j} - \underbrace{(\gamma_{0i} + \gamma_{1i}RFR \cdot P_j)}_{\text{moving costs from } j} + \epsilon_{ijk}$$

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- which is the same as the simple regression equation we began with
 - except for replacing max neighborhood with Emax neighborhood

Model and Estimation

- Want to estimate mobility and welfare under counterfactual *RFR*
- Simple analysis: Hold v and P constant under counterfactual *RFR*.
 - No assumptions/parameterization required for v and P
- GE Analysis: Equilibrium v and P affected by counterfactual *RFR*.
 - Requires assumptions/parameterization required for v and P
 - Need to estimate Supply

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(Somewhat Preliminary) Results

- Allow heterogeneous effects
 - Age, race, marital status, income, school aged kid
 - Results for white households only
- Add additional neighborhood controls
 - Address misspecification
 - MSA Fixed effects
 - PUMA characteristics

Results - Moving Cost Determinants

Variable	Estimate
Income	-0.34
Marriage Status	-0.48
School aged child	-0.22
Realtor Fee	-6.63
Income x Realtor Fee	2.11
Mean Moving Cost (\$100k)	0.56

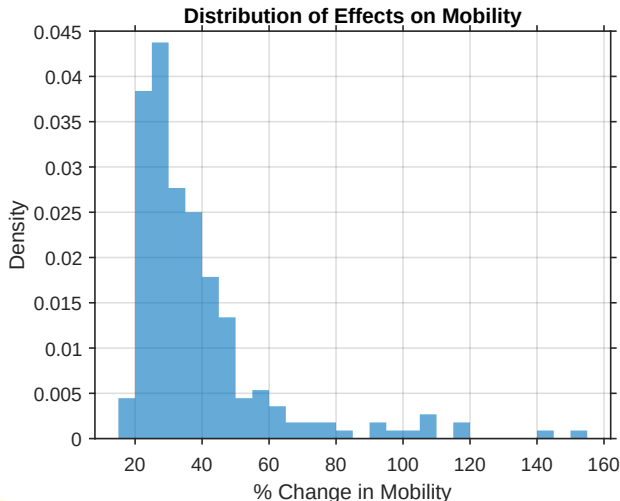
Counterfactual Analysis

- We consider a change in the RFR from 6% to 3%
- Simple analysis: Hold v and P constant
- Calculate individual specific % change in mobility
 - Report MSA-level means
 - Proxy for consumer welfare gains
- Calculate MSA-specific % change in total commissions
 - Likely an upper bound on Realtor welfare given low barriers to entry
- Calculate *RFR* that maximizes total commissions.
 - Peak of Laffer curve

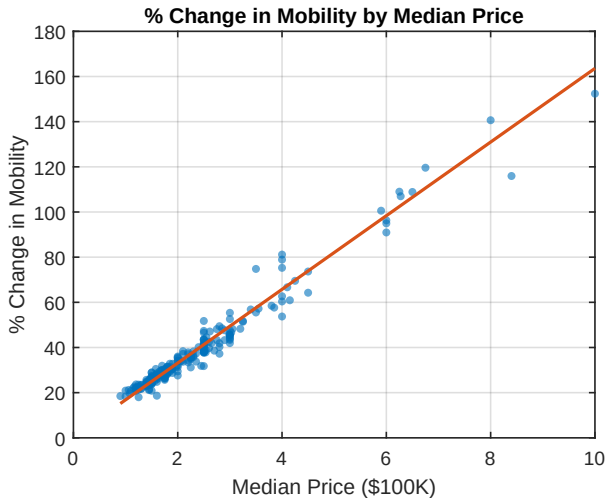
Counterfactual - Halving the Realtor Fee

MSA	% Change in Mobility	% Change in Realtor Commissions
Atlanta	38%	-29%
Chicago	42%	-26%
Dallas	38%	-28%
LA	119%	22%
New York	79%	-1%
Philadelphia	42%	-27%
Phoenix	43%	-26%

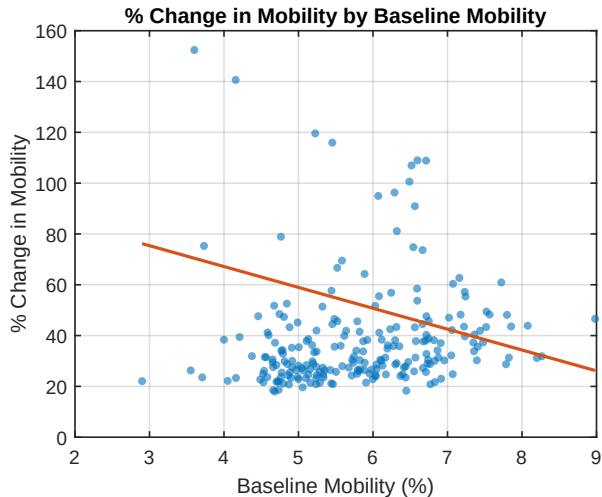
Counterfactual - Halving the Realtor Fee: Mean = 43%



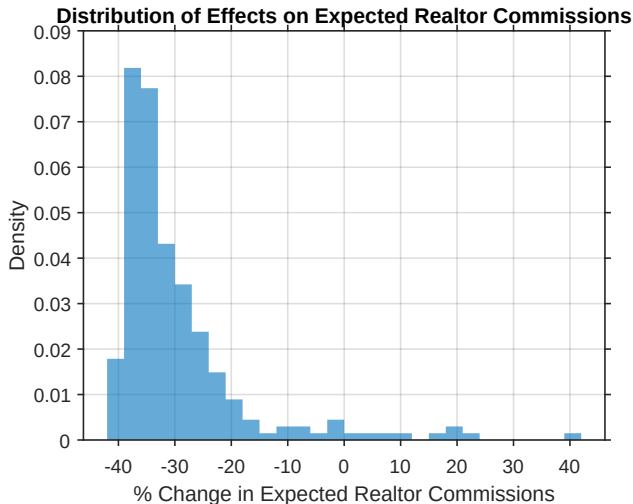
Counterfactual - Halving the Realtor Fee



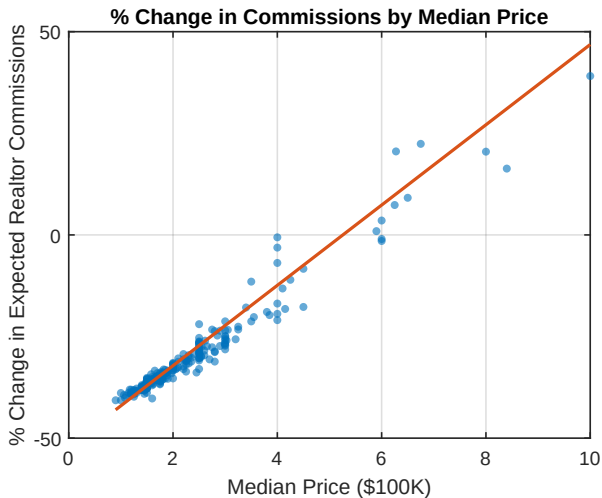
Counterfactual - Halving the Realtor Fee



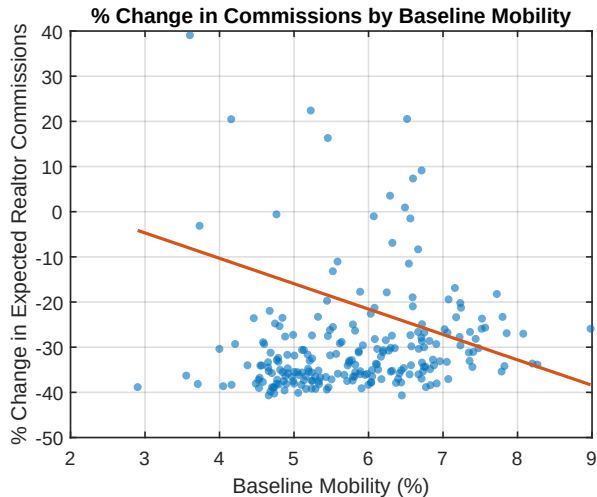
Counterfactual - Halving the Realtor Fee: Mean = -20%



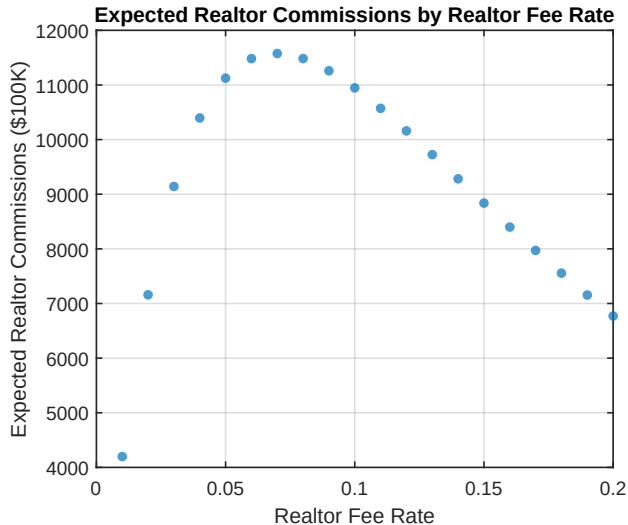
Counterfactual - Halving the Realtor Fee



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Counterfactual - Halving the Realtor Fee



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Conclusion

- Economic impacts of mobility costs are potentially sizeable.
- Realtor fees are large
- We develop a novel identification strategy to estimate effects of mobility costs
- Effects are estimated to be large
- Reducing Realtor fees by half would increase mobility by 43%
- Reducing Realtor fees by half would reduce total commissions by 20%

Mobility Costs – Realtor Fees

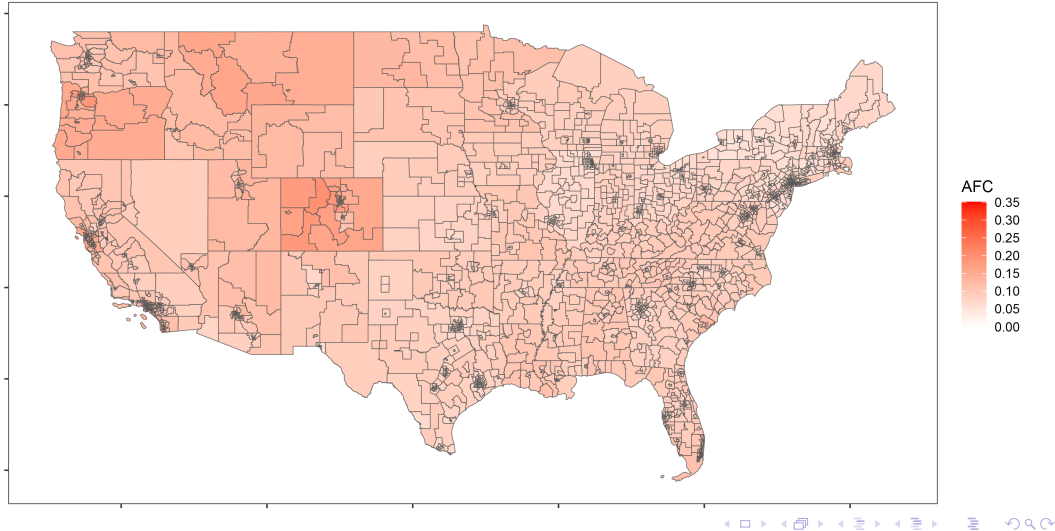
- Standard way to annualize costs = $UCR \cdot \text{Price}$
 - UCR denotes user-cost rate
- New way to view size of Realtor fee: Annualized Fraction of Cost

$$AFC = \frac{\text{Annualized Realtor Rate}}{UCR + \text{Annualized Realtor Rate}}$$

- Calculate at means: Realtor Rate, 6%; Tenure, 8 years; UCR 4.35%

$$AFC = \frac{6/8}{4.35 + 6/8} = 14.7\%$$

Mobility Costs – AFC – Holding Tenure Constant



Mobility Costs – AFC – Spatially Varying Tenure

